

Real-Time Fraud Detection System

Difficulty	Very Hard
Estimated Duration	20-25 Days
Best Dataset Link	https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

Project Workflow

- 1 Collect banking transaction dataset
- 2 Handle imbalanced data
- 3 Clean missing and duplicate records
- 4 Perform feature engineering
- 5 Train fraud detection model
- 6 Create real-time prediction API
- 7 Generate fraud alerts dashboard
- 8 Deploy Flask application

Company Real-World Workflow

- 1 Bank receives transactions
- 2 AI checks suspicious activity
- 3 Risk score generated
- 4 Fraud alert sent to security team

Tools & Technologies Used

- Python
- XGBoost
- Random Forest
- Flask
- Pandas
- NumPy
- Matplotlib

Data Cleaning Process

- Remove duplicate records
- Handle missing values
- Normalize and standardize data
- Remove outliers if required
- Feature engineering and transformation
- Train-test split preparation

Deployment & Final Output

Final model can be deployed using Flask, Streamlit, or FastAPI. Dashboard can be hosted on Render, Railway, or AWS cloud platforms.